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FLOODING PREDICTION IN A LARGE WATERSHED: AN EXAMPLE FROM THE MOHAWK RIVER IN NEW YORK

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Introduction

Increase in flooding in the rivers due to the climatic change has shown that natural hazards such as ice jams and tropical storms may result to flooding events in the rivers more often [1-5]. Ice jams occur when the frozen river breaks up and movement of large ice chunks is restricted at channel constrictions, lock stations (a fixed chamber where water level can be varied), and bridges. In this paper, we present a methodology in application to Mohawk River, and it can be applied at different USGS monitoring stations in the Mohawk watershed in Upstate New York, USA (Figure 1) or in other watersheds as well. For any watershed including the Mohawk watershed, it is important to know the time and the damage extent area of a possible flood event. The determination of the trigger level of flooding and the initiation time of flooding would predict chronic jam points or flooding events and begin to model what may happen as ice jams or tropical storms contribute to river flooding. A better understanding of flooding may reduce the chance of costly damages associated with these hazards.

Methodology

A methodology is described for the determination of the initial elevation of the Mohawk River where flooding may occur (Flood Trigger Level; FTL) as well as the initial time period that the flooding may occur in the river (Flood Initiation Time; FIT). The FTL has been determined using simulations of flooding with Geographical Information System software (GIS) and Light Detection and Ranging data (LiDAR). The FIT has been determined using the decomposition of the time series of the water discharge of the river, the ground water level, and the climatic variables. Once the river water level and the water discharge association has been established, then the two models can be integrated in one for each station of Mohawk River in order to derive a time-spatial model for the prediction of flooding in the river.

Determination of the Flood Trigger Level of the river

To estimate the Flood Trigger Level (FTL) of Mohawk River in different stations along the river, Air-LiDAR elevation data were used in order to construct a high-resolution Digital Elevation Model (DEM) of the specific station of Mohawk River. Due to variable slope and elevation along the river's longitudinal profile the river requires to be studied into segments (Fig. 1). FTL values should be determined on each segment along the Mohawk River using a previously described methodology [6].

At each segment all the contained elevation LiDAR points from a buffered area along the Mohawk River and the intersection of each of the 45 segments were extracted and were utilized to build a high resolution DEM. The extracted LiDAR points have been converted into a Triangulated Integrated Network (TIN). A polygon surface with an increasing elevation starting from the minimum available LiDAR elevation point in the river and using increments of less than a meter depending on the LiDAR vertical resolution the flooding progress can be simulated. This simulation provides through incremental water elevation levels a flooding surface and a volume change (Fig. 2). Simulation may produce high or low resolution of a map showing a flooding extent at different water level. This resolution depends on the size of the height increments (0.1 m) which is used for the incremental surface and volume calculations.

The TIN together with the elevation polygon surface is used in order to estimate the surface and the volume of the flooding area. Polygons overlaid for volumetric and surface calculations along the different segments require to be cut into at least 30 squares (fishnet) along a river length of 1 km. A greater amount of squares may result to more accurate determination of the FTL for the specific station. Figure 2 shows the flood trigger level for a specific segment of the Mohawk River. When the water level is higher than the bank full, then the water is spread over the flood plain. This methodology has been applied for the rest of the segments along the Mohawk River [7].

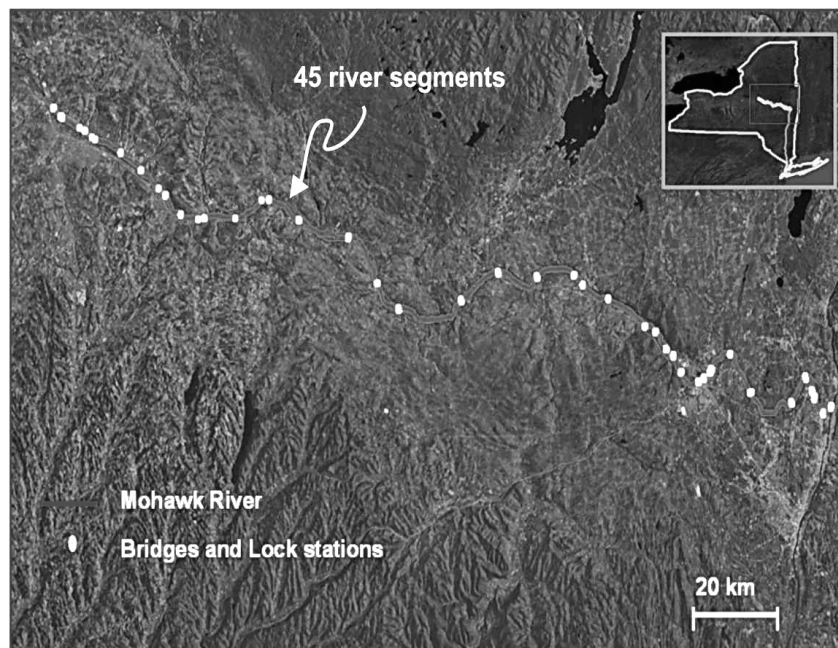


Figure 1. Mohawk River shown by red line has been studied along 45 segments constrained by natural or human structures (bridges and lock stations shown by white circles).

Determination of the Flooding Initiation Time of the River

Once we know the flood trigger level, we can apply a methodology for the estimation of the flooding initiation time event of the river using the water discharge time series. Figure 2 consists of a sequence of graphs that describe the association between the height gage and the water discharge using a polynomial or linear curve, depending on the flow level for the monitoring station at Schoharie Creek. Long-term discharge data from historical flood records including the most recent (February 2014-June 2014) continuous monitoring discharge versus height gage data shows a very high coefficient of determination, R^2 .

In particular, a polynomial association with $R^2=0.99$ between water discharge and height gage exists for the first 5,000 ft^3/sec . The flooding events showing greater values of discharge (5,000 - 128,000 ft^3/sec) can be described by a linear association with $R^2 = 0.99$. Water discharge may differ in different seasons. During the winter time and upon cold temperatures and ice presence the cross-sectional area of the river will decrease, and so the water discharge. This will require different prediction models for the winter and the summer incorporating temperature and precipitation records [8]. The water discharge and gage height association allows the connection of the determination of the FIT simulation model with the time series model for the prediction of the FIT. Therefore, we can predict the gage height of the river by the water discharge using Figure 2 for this specific river segment.

For the determination of the FIT of the river, we use a statistical methodology to decompose the time series of the water discharge time series, the ground water level, and the climatic variables (temperature, tides, wind speed and precipitation) into different components (long, seasonal and short term component). The long term component describes the fluctuations of a time series defined as being longer than a given threshold; the seasonal component describes the year-to-year fluctuations, while the short term component describes the short term variations. The main purpose is to predict the water discharge time series using the ground water level and the climatic variable for different stations in Mohawk River. For this reason, the Kolmogorov- Zurbenko (KZ) filter is used for the decomposition of the time series [9]. The KZ filter, which separates the long term variations from the short term variations in a time series, provides a simple design and the minimum level of interferences between the scales of the time series (long, seasonal and short term components).

In our analysis, we first decompose the time series of the water discharge derived by three different locations nearby the Mohawk River and then we predict each component of the water discharge time series

separately using a multivariate model. The decomposition of the time series provides a coefficient of determination of the water discharge up to 81%. Figure 3 shows some examples of the prediction of the water discharge time series for some of the monitoring stations in the Mohawk watershed in different years. This methodology can be applied in other stations, as well [10].

Conclusions

With the simulation model of the FTL determination using LiDAR data and GIS techniques and the decomposition of the time series model, we can estimate the FTL and the FIT for each station along Mohawk River. The design of multivariate models as well as the decomposition of the time series of the water discharge improves the prediction of flooding that can be caused by storms, rapid snowmelt and ice jams. Both models (spatial model using GIS and LiDAR and time series model) can be connected by estimating the association of the water discharge and height gage of the station. This methodology can be applied for real-time spatial analysis for disaster management or preventing a flooding hazard in a river such as in the Mohawk River in New York.

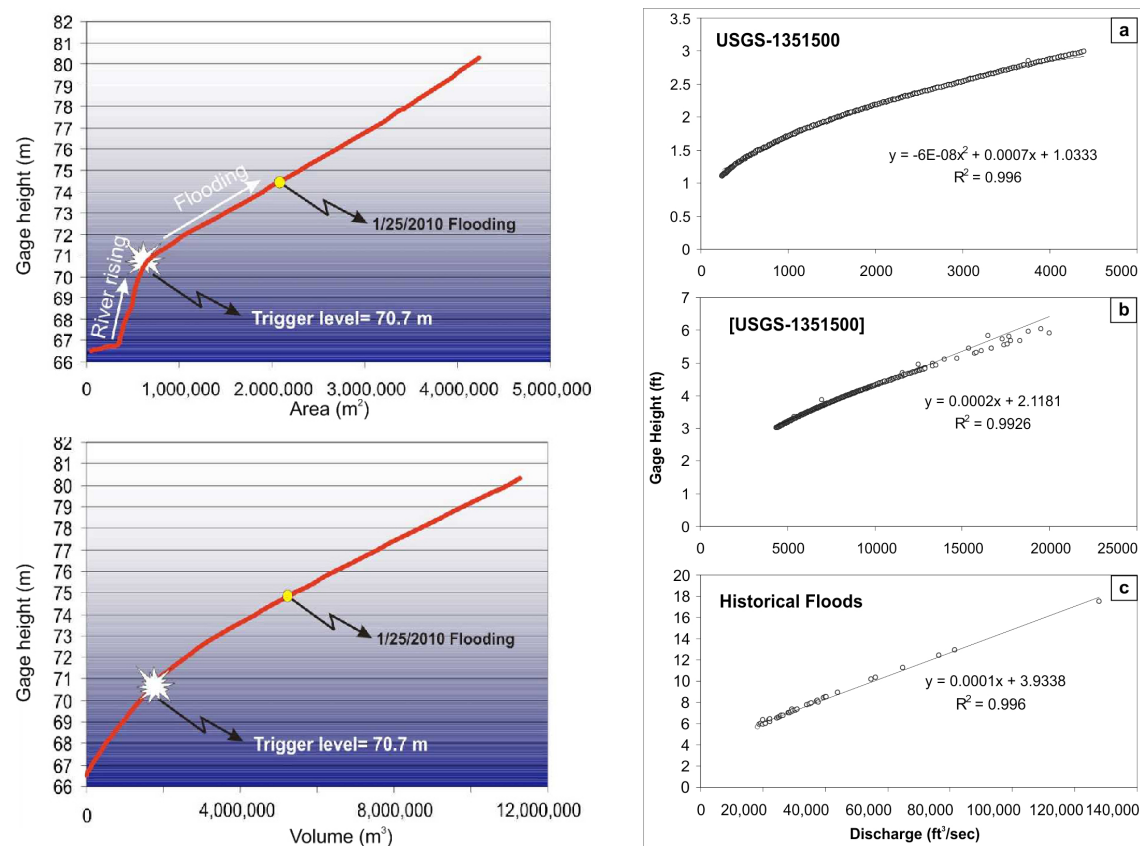


Figure 2. (Left) Flooding simulation from incremental (0.3 m) GIS volumetric and surface calculations [6]. (Right): Water discharge and height gage association in one of the USGS monitoring stations at Mohawk watershed (Schoharie Creek; USGS1351500). (a) Low flow extrapolation up to 5,000 ft³/sec of water discharge and the new gage height sensor from the continuous monitoring data (February of 2014 to June of 2014); (b) Medium flow extrapolation with some historical flood records; (c) historical flood records between 1940-2013.

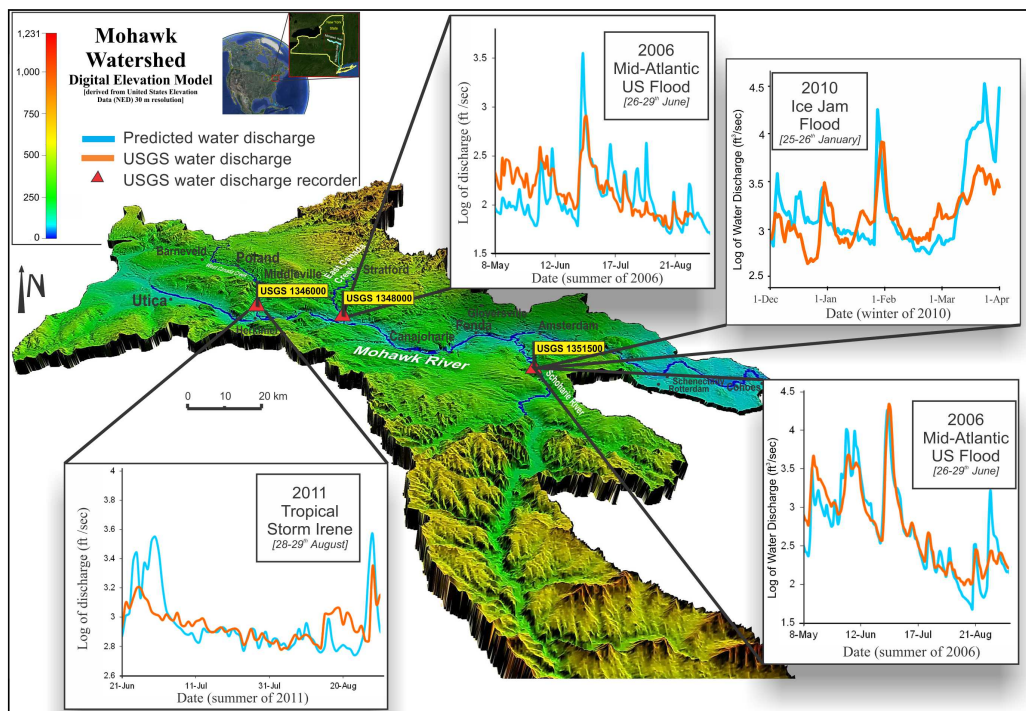


Figure 3. 3D Digital Elevation Model (DEM) of the Mohawk watershed in New York State, USA. Red triangles show the USGS stations where time series and GIS flooding prediction model has been applied. In the graphs, blue lines show the predicted water discharge values and orange lines show the USGS real data.

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Poster Presentation